

1. MailOrderHouse

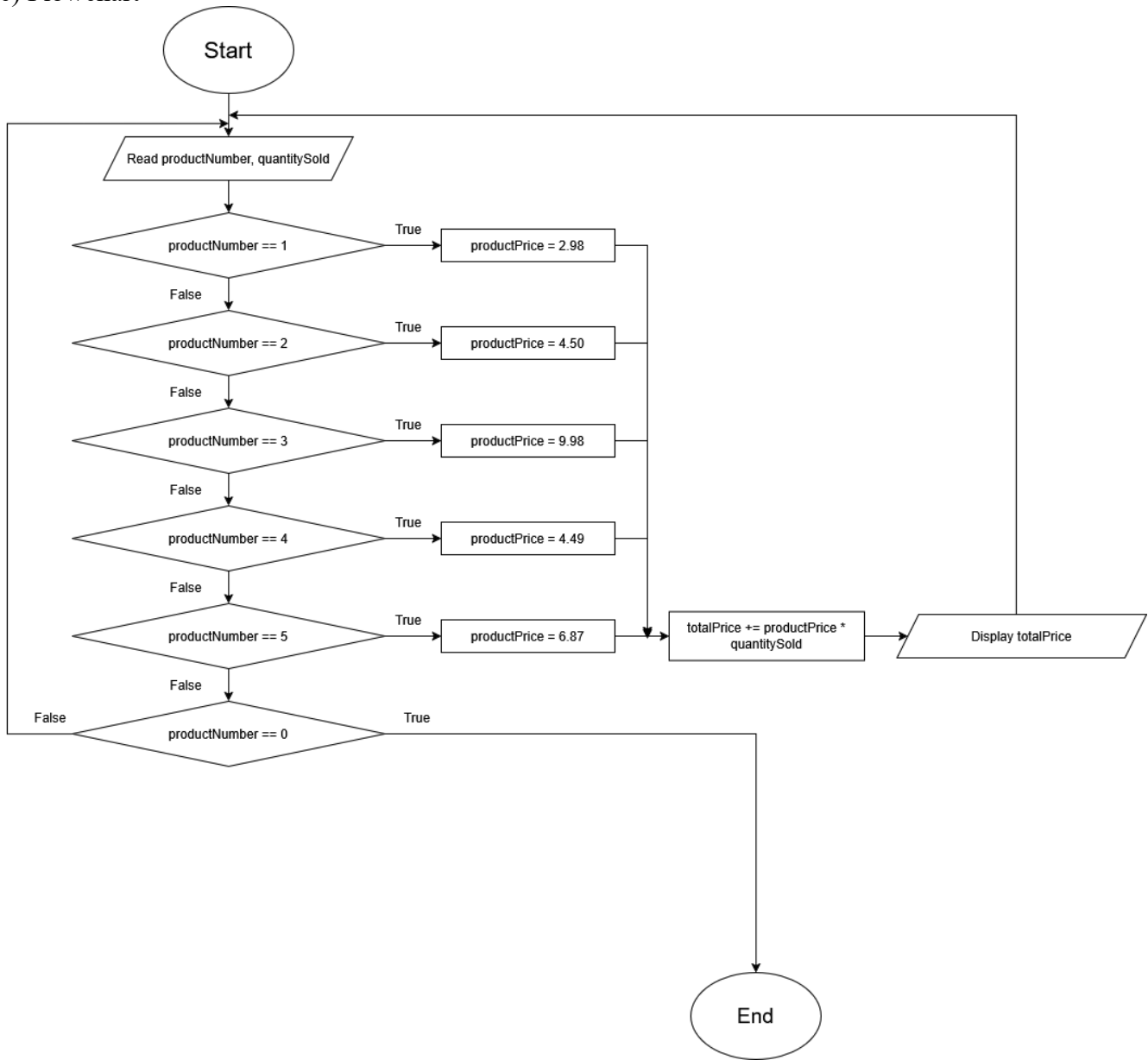
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Product number 2. Quantity sold	1. Total retail value of all products sold
Processing Required	Solution Alternative
Read productNumber, quantitySold switch(productNumber){ case 1-5: set productPrice case 0: exit } totalPrice += productPrice * quantitySold Display totalPrice Loop	1. Define product number and quantity sold as constants. 2. Define product number and quantity sold as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Product number 2. Quantity sold	1. Enter productNumber, quantitySold 2. switch(productNumber){ case '1': productPrice = 2.98;break; case '2': productPrice = 4.50;break; case '3': productPrice = 9.98;break; case '4': productPrice = 4.49;break; case '5': productPrice = 6.87;break; case '0': jump 6 default: Display "Please enter again." jump 1 } 3. totalPrice += productPrice * quantitySold 4. Display totalPrice 5. Jump 1 6. End	-	1. Total retail value of all products sold

c) Flowchart



d) Pseudocode

Start

```

while(true){
    Read productNumber, quantitySold
    switch(productNumber){
        case '1':
            productPrice = 2.98; break;
        case '2':
            productPrice = 4.50; break;
        case '3':
            productPrice = 9.98; break;
        case '4':
            productPrice = 4.49; break;
    }
    totalPrice += productPrice * quantitySold
    Display totalPrice
}

```

```

        case '5':
            productPrice = 6.87; break;
        case '0':
            System.exit(0);
        default:
            print "Please enter again."
            skip = true
    }
    if (!skip) {
        totalPrice += productPrice * quantitySold
        Display totalPrice
    }
}

```

End

e) Java code

```

import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 6
public class MailOrderHouse {
    public static void main(String[] args) {
        System.out.println("Welcome to mail-order house!");
        double productPrice = 0, totalPrice = 0;
        char input;
        int quantitySold = 0;
        Scanner scanner = new Scanner(System.in);
        boolean skip = false;

        while (true) {
            System.out.println("\nPlease enter the product number (1 to 5) and the quantity sold." +
                               "\nEnter 0 to exit the program.");
            skip = false;

            System.out.println("Product Number : ");
            input = scanner.next().charAt(0);

            switch (input) {
                case '0':
                    System.out.println("Total Price : RM" + String.format("%.2f", totalPrice));
                    System.exit(0);
                case '1':
                    productPrice = 2.98;
                    break;
                case '2':
                    productPrice = 4.50;
                    break;
                case '3':
                    productPrice = 9.98;
                    break;
                case '4':
                    productPrice = 4.49;
                    break;
                case '5':
                    productPrice = 6.87;
                    break;
                default:
                    System.out.println("Please enter a correct number.");
                    scanner.nextLine();
                    skip = true;
            }

            if (!skip) {
                System.out.println("Quantity Sold : ");
                try {
                    quantitySold = scanner.nextInt();
                } catch (InputMismatchException e) {
                    System.out.println("Please enter a integer.");
                    scanner.nextLine();
                    skip = true;
                }
                if (quantitySold < 0) {
                    System.out.println("Please enter a positive integer.");
                    scanner.nextLine();
                    skip = true;
                }
            }

            if (!skip) {
                totalPrice += productPrice * quantitySold;
                System.out.println("Total Price : RM" + String.format("%.2f", totalPrice));
            }
        }
    }
}

```

2. ZooNegaraFeeCalculator

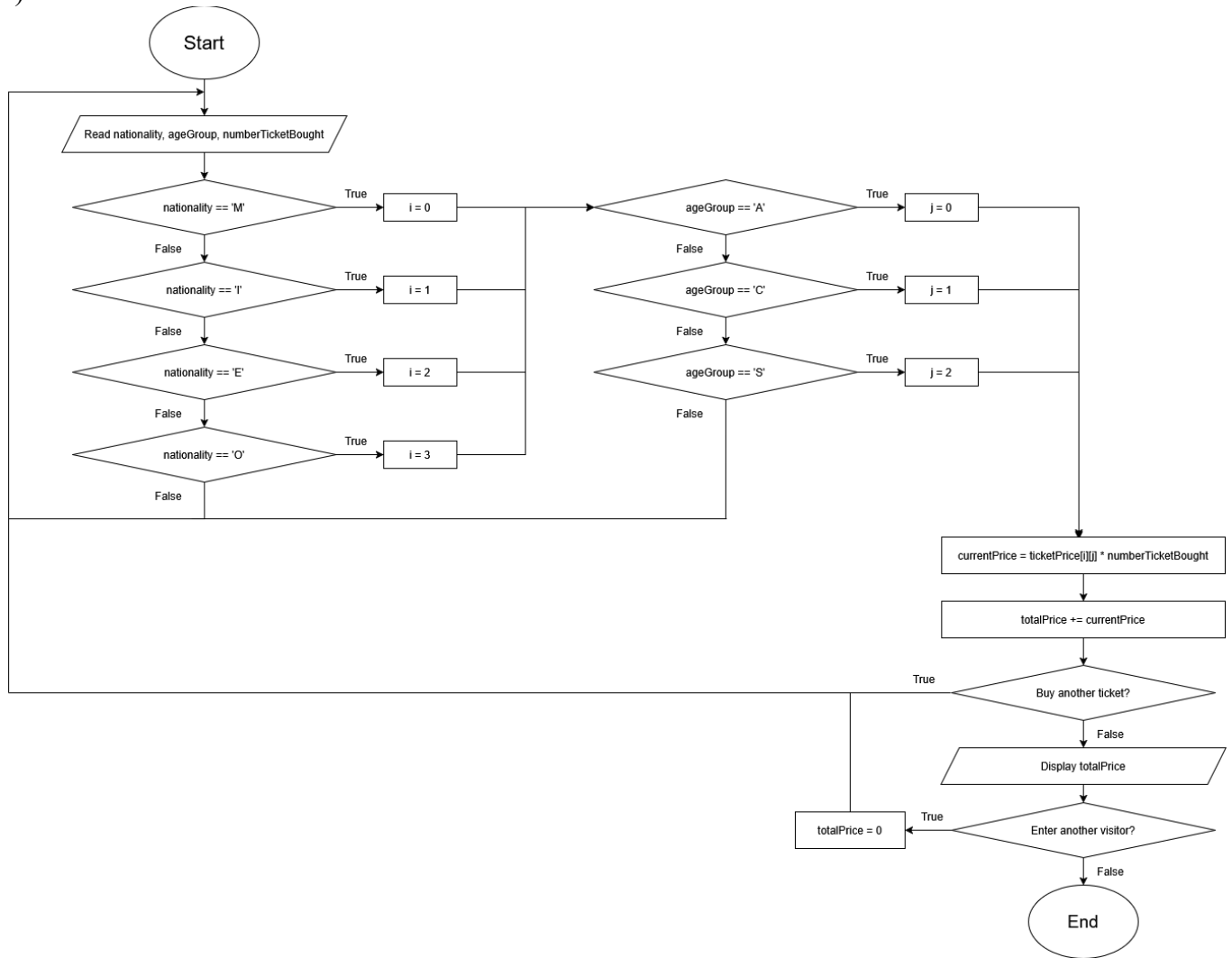
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Nationality 2. Age group 3. Number of tickets bought	1. Total ticket price
Processing Required	Solution Alternative
<pre> Read nationality, ageGroup, numberTicketBought final int[][] ticketPrice = {{43,16,21},{50,25,50}, {61,33,61},{82,43,82}}; int i,j; switch(nationality){ case 'M': i = 0; break; case 'T': i = 1; break; case 'E': i = 2; break; case 'O': i = 3; break; } switch(ageGroup){ case 'A': j = 0; break; case 'C': j = 1; break; case 'S': j = 2; break; } currentPrice = ticketPrice[i][j] * numberTicketBought totalPrice += currentPrice Ask another ticket Display totalPrice Ask another visitor </pre>	1. Define nationality, age group, number of tickets bought as constants. 2. Define nationality, age group, number of tickets bought as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Nationality 2. Age group 3. Number of tickets bought	<pre> 1. Enter nationality, ageGroup, numberTicketBought 2. Set ticketPrice[][] 3. Set int i,j 4. switch(nationality){ case 'M': i = 0; break; case 'T': i = 1; break; case 'E': i = 2; break; case 'O': i = 3; break; } 5. switch(ageGroup){ case 'A': j = 0; break; case 'C': j = 1; break; case 'S': j = 2; break; } 6. currentPrice = ticketPrice[i][j] * numberTicketBought 7. totalPrice += currentPrice 8. Display "Buy other ticket (Y/N)?:" 9. if ((input == 'Y') (input == 'y')){Jump 1} 10. Display totalPrice 11. Display "Enter another visitor (Y/N)?:" 12. if ((input == 'Y') (input == 'y')){Display "*****VISITOR" + indexVisitor+ "*****"; indexVisitor++;totalPrice = 0;} 13. End </pre>	-	1. Total ticket price

c) Flowchart



d) Pseudocode

Start

```

hasAnotherVisitor = true;
final int[][] ticketPrice = {{43,16,21},{50,25,50},{61,33,61},{82,43,82}};
for(int indexVisitor = 1; hasAnotherVisitor ; indexVisitor++, totalPrice = 0){
    while(hasAnotherTicket){
        Read nationality, ageGroup, numberTicketBought
        switch(nationality){
            case 'M':
                i = 0; break;
            case 'I':
                i = 1; break;
            case 'E':
                i = 2; break;
            case 'O':
                i = 3; break;
        }
    }
}
  
```

```

        switch(ageGroup){
            case 'A':
                j = 0; break;
            case 'C':
                j = 1; break;
            case 'S':
                j = 2; break;
        }
        currentPrice = ticketPrice[i][j] * numberTicketBought;
        totalPrice += currentPrice;
        Read hasAnotherTicket
    }
    Display totalPrice
    Read hasAnotherVisitor
}

End

```

e) Java code

```

import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 6 Q2
public class ZooNegaraFeeCalculator{

    public static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args){
        System.out.println("Welcome to the Zoo Negara!\nEnter the ticket you want to buy!\n");

        final int[][] ticketPrice = {{43,16,21},{50,25,50},{61,33,61},{82,43,82}};

        int i,j = 0;
        int totalPrice = 0;

        boolean hasAnotherVisitor = true;

        // Loop for multiple visitor
        for(int indexVisitor = 1; hasAnotherVisitor ; indexVisitor++, totalPrice = 0){
            System.out.println("***** VISITOR " + indexVisitor + " *****\n");

            // Loop for multiple ticket
            do{
                int numberTicketBought = 0;

                // Get all required infomation about the ticket from user
                while(true){
                    char nationality = getCharResponse("Enter the nationality (M = Malaysian, I = Kad/Working Permit/Dependant Pass, E = ASEAN, O = Others) : ");
                    switch(nationality){
                        case 'M':
                            i = 0; break;
                        case 'I':
                            i = 1; break;
                        case 'E':
                            i = 2; break;
                        case 'O':
                            i = 3; break;
                        default:
                            System.out.println("Bad response for nationality!");
                            continue;
                    }
                }

                char ageGroup = getCharResponse("Enter the age group (A = Adult, C = Children (3 to 12 years old), S = Senior (60 years and above)) : ");
            } while(numberTicketBought > 0);
        }
    }
}

```

```

        switch(ageGroup){
            case 'A':
                j = 0; break;
            case 'C':
                j = 1; break;
            case 'S':
                j = 2; break;
            default:
                System.out.println("Bad response for age group!");
                continue;
        }

        numberTicketBought = getIntResponse("Enter the number of ticket to buy : ");
        if(numberTicketBought < 0){
            System.out.println("Bad response for number of ticket!");
            continue;
        }

        break;
    }

    // Calculate ticket price
    int currentPrice = ticketPrice[i][j] * numberTicketBought;
    System.out.println("Ticket/Tickets for RM" + String.format("%.2f", (double)currentPrice) + " is/are charged.\n");
    totalPrice += currentPrice;

    } while(getBooleanResponse("Buy other ticket? (Y/N) : ")); // Ask for another ticket

    // Display total price
    System.out.println("\nThe total payment for VISITOR " + indexVisitor + " is RM" + String.format("%.2f", (double)totalPrice) + "\n");

    // Ask for another visitor
    hasAnotherVisitor = getBooleanResponse("Enter another visitor (Y/N) : ");
    System.out.println("");
}

System.out.println("Thanks for using the system!");
}

public static char getCharResponse(String question){
    System.out.println(question);
    return scanner.next().charAt(0);
}

public static int getIntResponse(String question){
    int input;
    while(true){
        System.out.println(question);
        try{
            input = scanner.nextInt();
        } catch (InputMismatchException e){
            System.out.println("Bad response! Please enter again.");
            scanner.nextLine();
            continue;
        }
        break;
    }
    return input;
}

public static boolean getBooleanResponse(String question){
    boolean input;
    char response;
    while(true){
        System.out.println(question);
        response = scanner.next().charAt(0);
        if(response == 'Y' || response == 'y') input = true;
        else if(response == 'N' || response == 'n') input = false;
        else {
            System.out.println("Bad response! Please enter again.");
            continue;
        }
        break;
    }
    return input;
}
}

```


3. HealthCalculator

a) Problem Analysis Chart (PAC)

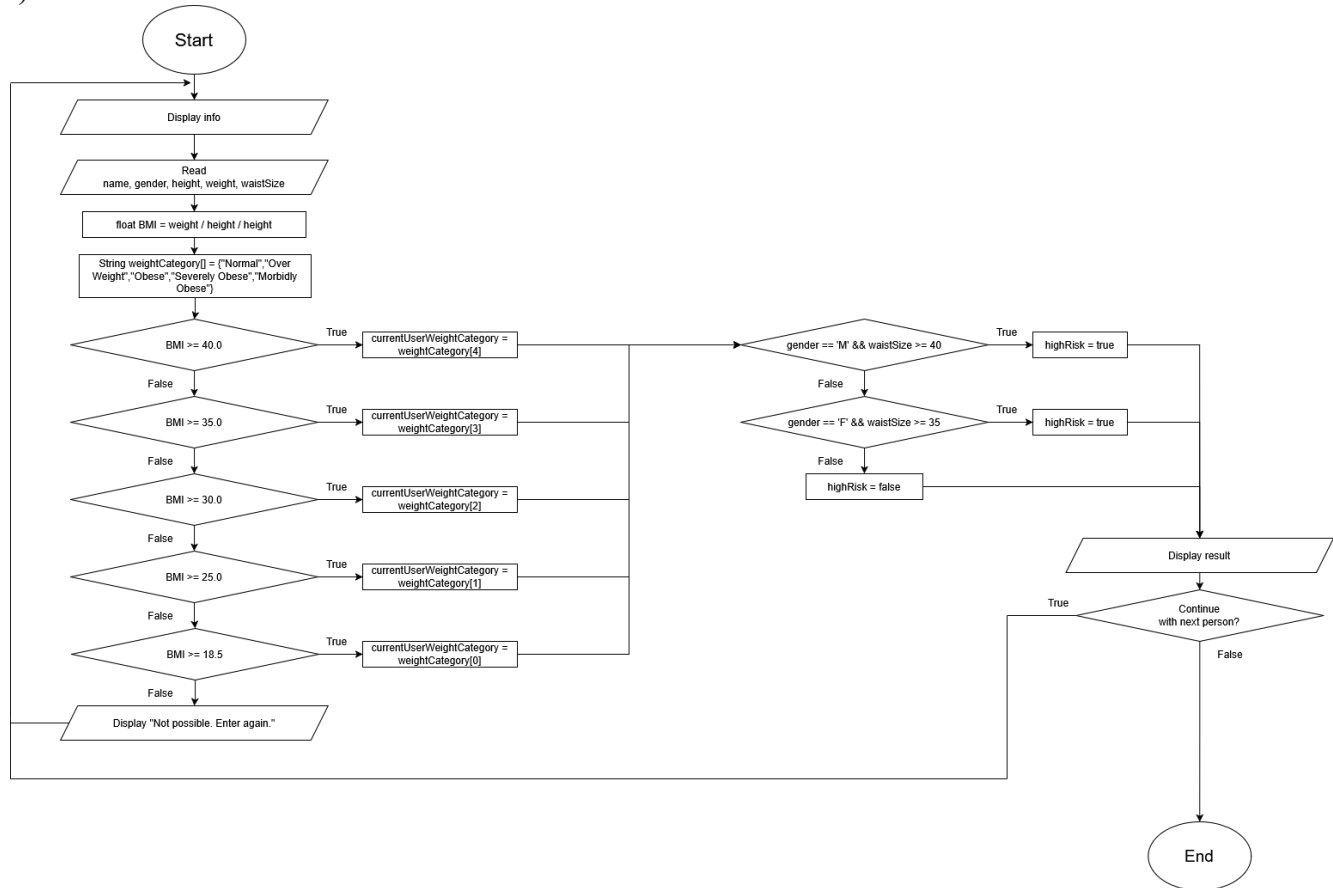
Given Data	Required Results
1. Name 2. Gender 3. Height 4. Weight 5. Waist size	1. BMI 2. Weight categories 3. Risk for heart disease and type 2 diabetes
Processing Required	Solution Alternative
Display info Read name, gender, height, weight, waistSize $BMI = \text{weight} / \text{height} / \text{height}$ Categorise weight with BMI range with if-else case Check if has higher risk for heart disease and type 2 diabetes with the data of waist size and gender Display result Ask for continuing next person	1. Define name, gender, height, weight, waist size as constants. 2. Define name, gender, height, weight, waist size as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Name 2. Gender 3. Height 4. Weight 5. Waist size	1. Display info 2. Read name, gender, height, weight, waistSize 3. $\text{float BMI} = \text{weight} / \text{height} / \text{height}$ 4. $\text{String[]} \text{weightCategory} = \{ \text{"Normal"}, \text{"Over Weight"}, \text{"Obese"}, \text{"Severely Obese"}, \text{"Morbidly Obese"} \}$ 4. if (BMI >= 40.0) currentUserWeightCategory = weightCategory[4] else if (BMI >= 35.0) currentUserWeightCategory = weightCategory[3] else if (BMI >= 30.0) currentUserWeightCategory = weightCategory[2] else if (BMI >= 25.0) currentUserWeightCategory = weightCategory[1] else if (BMI >= 18.5) currentUserWeightCategory = weightCategory[0] else throw error 5. if (gender == 'M' && waistSize >= 40) highRisk = true else if (gender == 'F' && waistSize >= 35) highRisk = true else highRisk = false	-	1. BMI 2. Weight categories 3. Risk for heart disease and type 2 diabetes

	6. Display result 7. Display "Continue with next person?" 8. if ((input == 'Y') (input == 'y')) Jump 1 9. End		
--	---	--	--

c) Flowchart



d) Pseudocode

Start

Display info

Read name, gender, height, weight, waistSize

float BMI = weight / height / height

String[] weightCategory = {"Normal", "Over Weight", "Obese", "Severely Obese", "Morbidly Obese"}

if (BMI >= 40.0)

 currentUserWeightCategory = weightCategory[4]

else if (BMI >= 35.0)

 currentUserWeightCategory = weightCategory[3]

else if (BMI >= 30.0)

 currentUserWeightCategory = weightCategory[2]

else if (BMI >= 25.0)

 currentUserWeightCategory = weightCategory[1]

else if (BMI >= 18.5)

 currentUserWeightCategory = weightCategory[0]

```
else
    Ask user enter again
    if yes, jump to the start

if (gender == 'M' && waistSize >= 40)
    highRisk = true
else if (gender == 'F' && waistSize >= 35)
    highRisk = true
else
    highRisk = false
Display result
Display "Continue with next person?"
if ((input == 'Y') || (input == 'y')) Jump to the start
```

End

e) Java code

```
import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 6 Q3
public class HealthCalculator{

    public static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args){
        String[] weightCategory = {"Normal","Over Weight","Obese","Severely Obese","Morbidly Obese","Undefined"};
        String name;
        float height,weight,waistSize;
        boolean isMale = false;
        String currentUserWeightCategory;
        boolean isHighRisk = false;

        do{
            // Display general info
            System.out.println("BMI information: \n"
                + "BMI (18.5-24.9) → Normal \n"
                + "BMI (25-29.9) → Over Weight \n"
                + "BMI (30-34.9) → Obese \n"
                + "BMI (35-39.9) → Severely Obese \n"
                + "BMI (≥40) → Morbidly Obese \n"
                + "***** \n"
                + "CAUTION!! \n"
                + "Women : waist size > 35 inches \n"
                + "Men : waist size > 40 inches \n"
                + "—— HIGHER RISK FOR HEART DISEASE AND TYPE 2 DIABETES!! —— \n"
                + "***** \n"
                + "\n"
                + "HEALTH CALCULATOR \n");

            // Get all required information about the user
            while(true){
                name = getStringResponse("Enter your name : ");

                char gender = getCharResponse("Enter your gender (M/F) : ");
                if(gender == 'M' || gender == 'm') isMale = true;
                else if (gender == 'F' || gender == 'f') isMale = false;
                else {
                    System.out.println("Bad response for gender!");
                    continue;
                }

                height = getFloatResponse("Enter your height in meters : ");
                if(height < 0) {
                    System.out.println("Bad response for height!");
                    continue;
                }

                weight = getFloatResponse("Enter your weight in kilogram : ");
                if(weight < 0) {
                    System.out.println("Bad response for weight!");
                    continue;
                }

                waistSize = getFloatResponse("Enter your waist size in inches : ");
                if(waistSize < 0) {
                    System.out.println("Bad response for waist size!");
                    continue;
                }

                break;
            }

            // Calculation
            float BMI = weight / height / height;
        }
    }
}
```

```

        if (BMI ≥ 40.0)
            currentUserWeightCategory = weightCategory[4];
        else if (BMI ≥ 35.0)
            currentUserWeightCategory = weightCategory[3];
        else if (BMI ≥ 30.0)
            currentUserWeightCategory = weightCategory[2];
        else if (BMI ≥ 25.0)
            currentUserWeightCategory = weightCategory[1];
        else if (BMI ≥ 18.5)
            currentUserWeightCategory = weightCategory[0];
        else
            currentUserWeightCategory = weightCategory[5];

        if (isMale && waistSize ≥ 40)
            isHighRisk = true;
        else if (!isMale && waistSize ≥ 35)
            isHighRisk = true;
        else
            isHighRisk = false;

        // Display result
        System.out.println(name + ", your BMI is " + String.format("%.2f", BMI) + ", so that means you are " + currentUserWeightCategory + ".");
        if (isHighRisk) System.out.println("You have higher risk for heart disease and diabetes.");

    } while (hasNextPerson()); // Ask for continue to next person

    System.out.println("Thank you for using the system.");

}

public static String getStringResponse(String question){
    System.out.println(question);
    String input = scanner.nextLine();
    return input;
}

public static char getCharResponse(String question){
    System.out.println(question);
    char input = scanner.next().charAt(0);
    scanner.nextLine();
    return input;
}

public static float getFloatResponse(String question){
    float input;
    while(true){
        System.out.println(question);
        try{
            input = scanner.nextFloat();
        } catch (InputMismatchException e){
            System.out.println("Bad response! Please enter again.");
            scanner.nextLine();
            continue;
        }
        break;
    }
    scanner.nextLine();
    return input;
}

public static boolean hasNextPerson(){
    boolean input;
    char response;
    while(true){
        System.out.println("Continue with next person?");
        response = scanner.next().charAt(0);
        if(response == 'Y' || response == 'y') input = true;
        else if(response == 'N' || response == 'n') input = false;
        else {
            System.out.println("Bad response! Please enter again.");
            continue;
        }
        break;
    }
    scanner.nextLine();
    return input;
}
}

```

4. MileageClaim

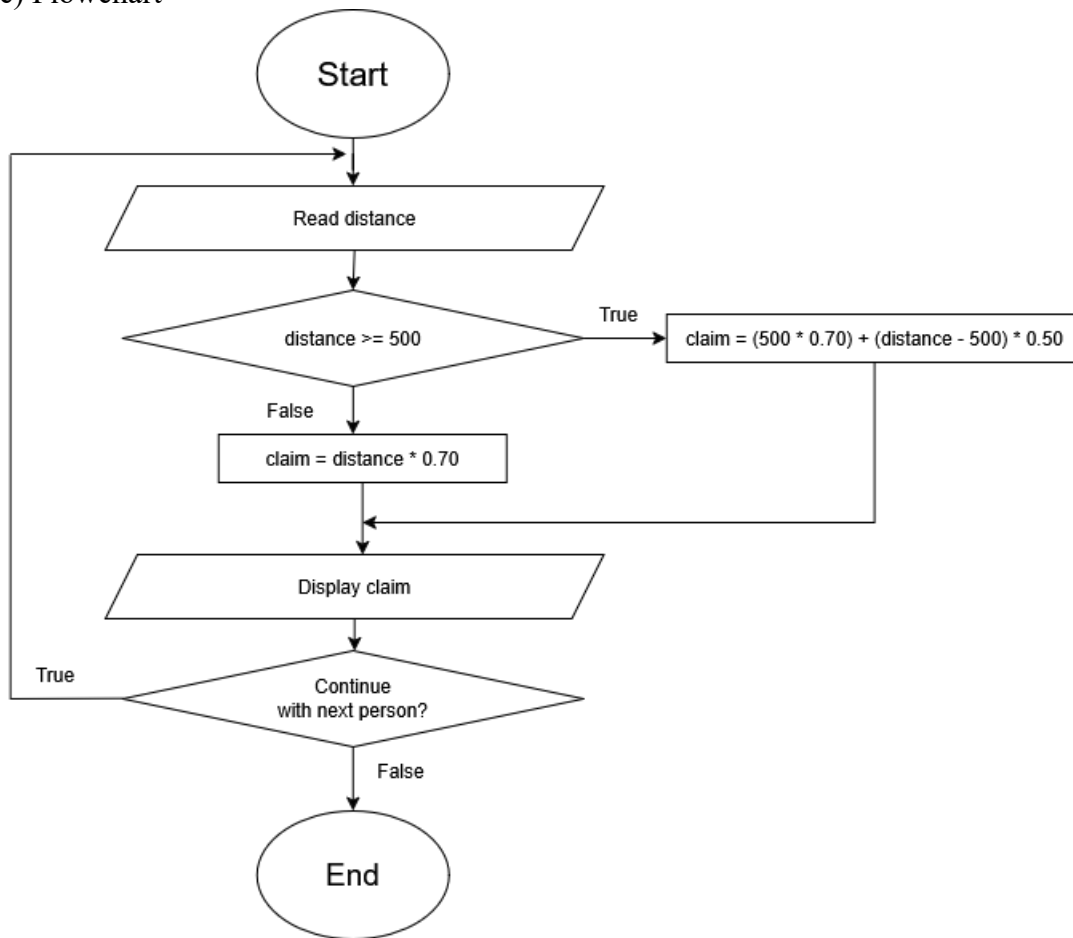
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Distance	1. Mileage Claim
Processing Required	Solution Alternative
Read distance if(distance >= 500){ claim = (500 * 0.70) + (distance - 500) * 0.50 } else { claim = distance * 0.70 } Display claim Ask for next staff	1. Define distance as constants. 2. Define distance as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Distance	1. Read distance 2. if(distance >= 500){ claim = (500 * 0.70) + (distance - 500) * 0.50 } else { claim = distance * 0.70 } 3. Display claim 4. Display "Continue with next staff?" 5. if ((input == 'Y') (input == 'y')) Jump 1 6. End	-	1. Mileage Claim

c) Flowchart



d) Pseudocode

Start

```
Read distance
if(distance >= 500){
    claim = (500 * 0.70) + (distance - 500) * 0.50
} else {
    claim = distance * 0.70
}
Display claim
Display "Continue with next staff?"
if ((input == 'Y') || (input == 'y')) {Jump to the start}
```

End

e) Java code

```

import java.util.Scanner;
import java.util.InputMismatchException;
// NGO YUK HANG 228991 LAB 6 Q4
public class MileageClaim{

    public static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args){
        do{
            double distance;

            while(true){
                System.out.println("Claim your mileage by inputing the distance (km) : ");
                try{
                    distance = scanner.nextDouble();
                } catch (InputMismatchException e){
                    System.out.println("Please input a number!");
                    continue;
                }
                break;
            }

            double claim;
            if(distance >= 500.0f){
                claim = (500 * 0.70) + ((distance - 500.0) * 0.50);
            }else{
                claim = distance * 0.70;
            }

            System.out.println("Your claim is RM" + String.format("%.2f",claim));
        } while(hasNextStaff());
    }

    public static boolean hasNextStaff(){
        boolean input;
        char response;
        while(true){
            System.out.println("Continue with next staff?");
            response = scanner.next().charAt(0);
            if(response == 'Y' || response == 'y') input = true;
            else if(response == 'N' || response == 'n') input = false;
            else {
                System.out.println("Bad response! Please enter again.");
                continue;
            }
            break;
        }
        scanner.nextLine();
        return input;
    }
}

```